

# \*Breeding of *Phalaenopsis*

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# \*Lecturer

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**<Education Background>**

**- Ph D. in Horticulture : from ChungBook National University**

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## 1. About *Phalaenopsis*

## 2. Breeding

A. Breeding purpose

B. Breeding technique - Cross breeding

C. Sterilization and sowing

D. Growing and transplanting

E. Process of the breeding

## 3. New varieties breeding by UNiPlantech

# \* 1. About Phalaenopsis

## \* Etymology :

- \* The name comes from Greek “Phalaina+Opisi” which means “A flower look like butterfly”
- \* The amount of original species : About 60 species

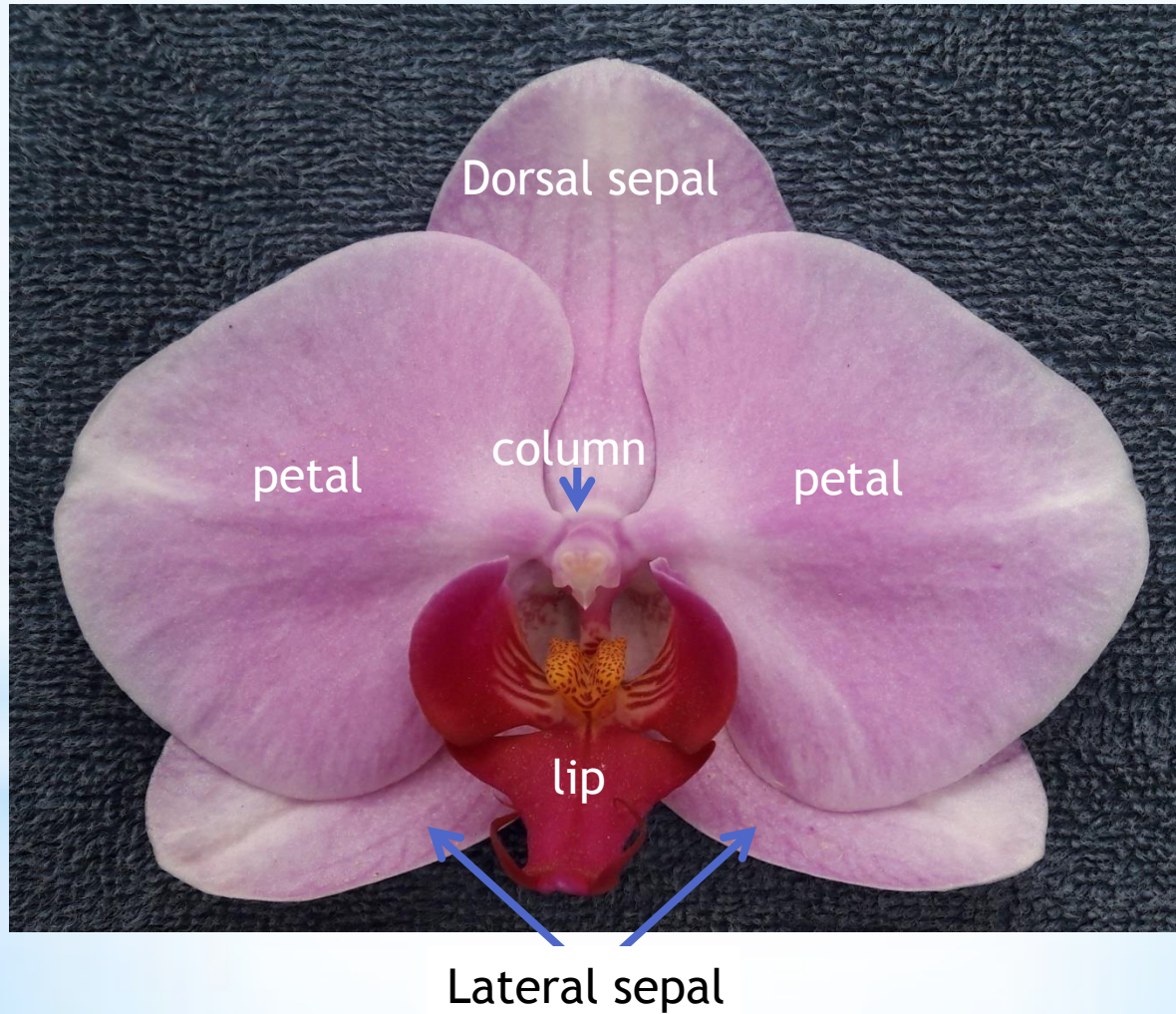
## \* Classification

- \* Orchidaceae(Orchid family) - Epidendroideae - Vandeae - Aeridinae - Phalaenopsis Blume

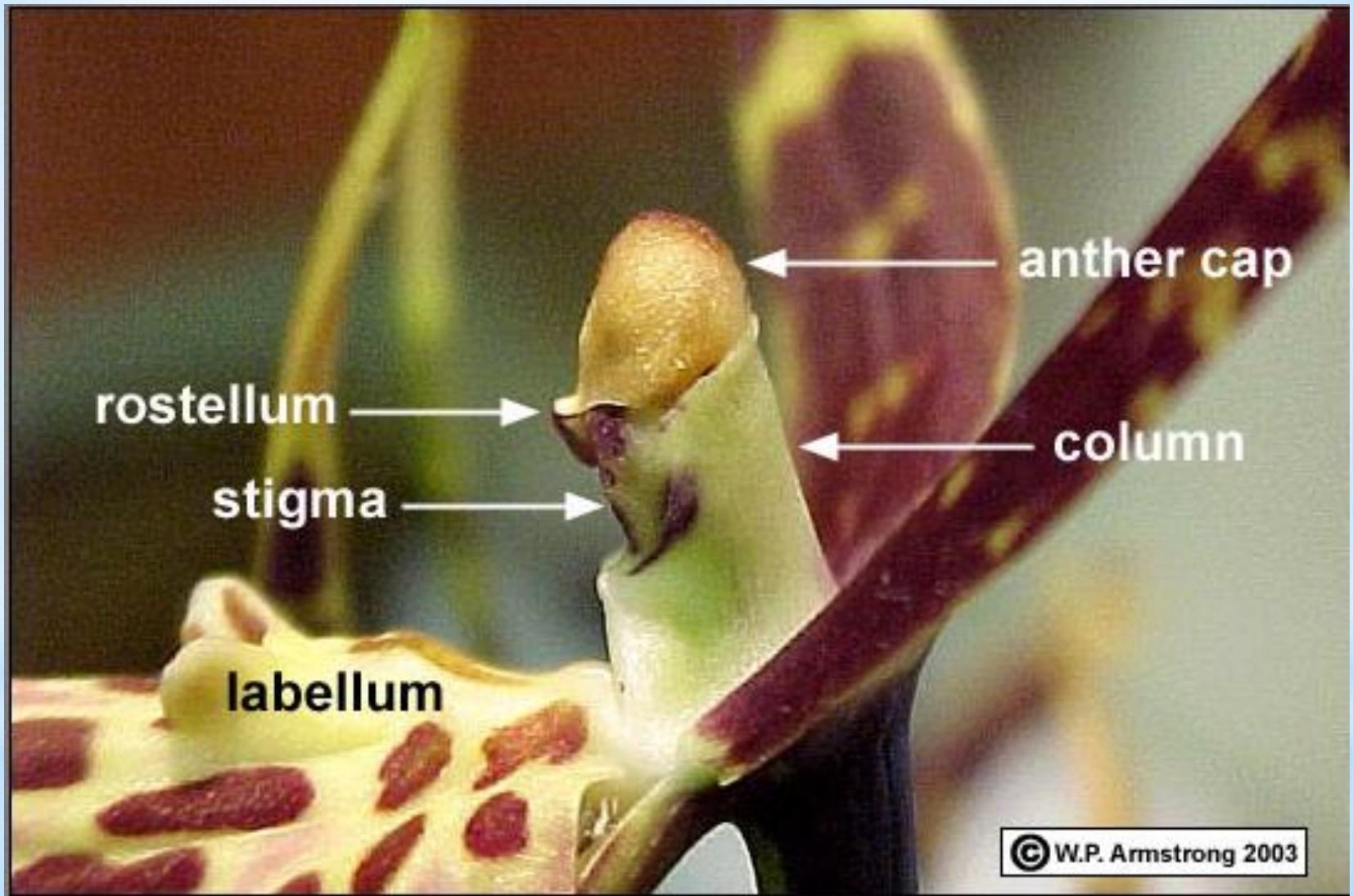
# \*The habitat

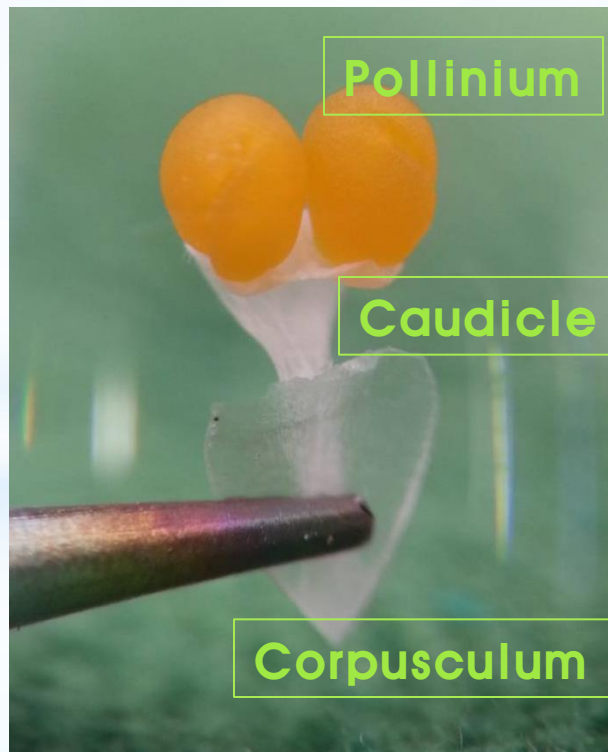
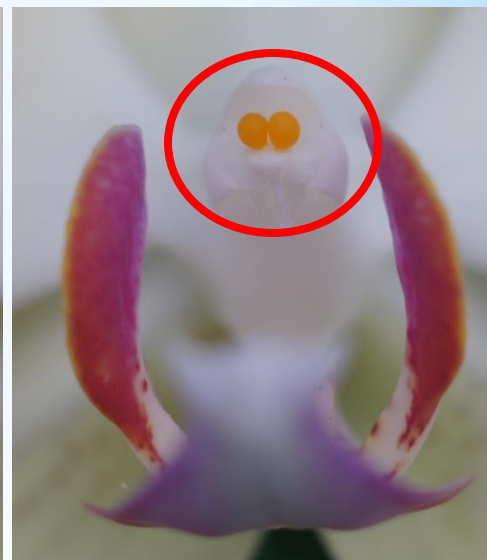
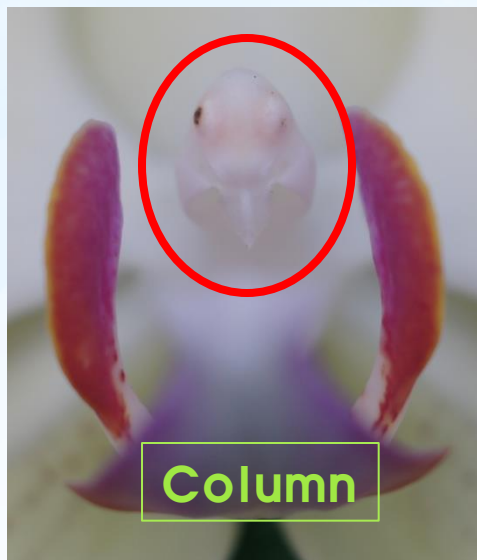
- \* Tropical Asia like Philippine, Malaysia, Indonesia, Vietnam, Taiwan and the north area of Australia
- \* In nature, *Phalaenopsis* species are typically fond of warm temperatures, thriving in temperatures around 20 to 35 °C (68-95 °F), but are adaptable to conditions more comfortable for human habitation in temperate zones (15 to 30 °C or 59 to 86 °F).
- \* *Phalaenopsis* requires high humidity (60-70%) and low light of 12,000 to 20,000 lux.
- \* However, *Phalaenopsis* orchids can adapt to the lower humidity found in most homes.

# \* The Flower structure of Phalaenopsis











## \* 2. Breeding

- \* *Phalaenopsis* are among the most popular orchids sold as potted plants, owing to the ease of propagation and flowering under artificial conditions.
- \* Although the number of flower, flower size and color are important purpose to improve varieties, it is different to the taste, type of people and it has been changing according to the time, nation and area.
- \* But the below characteristics (like a strength and thickness of peduncle, a strength of pedicle, an arrangement of flowers etc..) are unchanging important breeding goal

## \* A. Breeding purpose

### 1. Flower characters :

- size, number, color, shape, unique, flowering life, fragrances

### 2. Plant characteristics :

- shape of leaf, color, up light and down etc...

### 3. Genetic stability : stable from somatic mutation

### 4. Cultivation characteristics :

- Easy to in vitro propagation
- Easy to growing(growing rate-speed, resistance of disease...)
- Easy to flowering control(flowering time, response of low temperature, falling flower bud...)

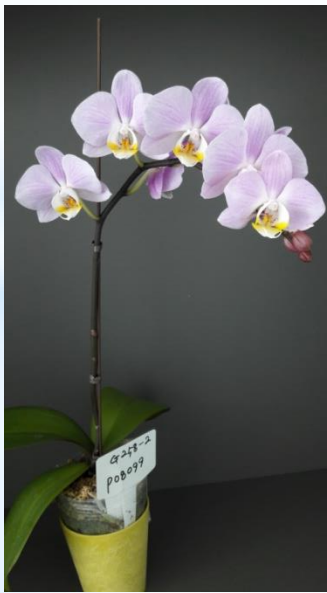
# \* 1. Flower characters :



size, number, color, shape, unique, flowering life, fragrances



## \* 2. Plant characteristics

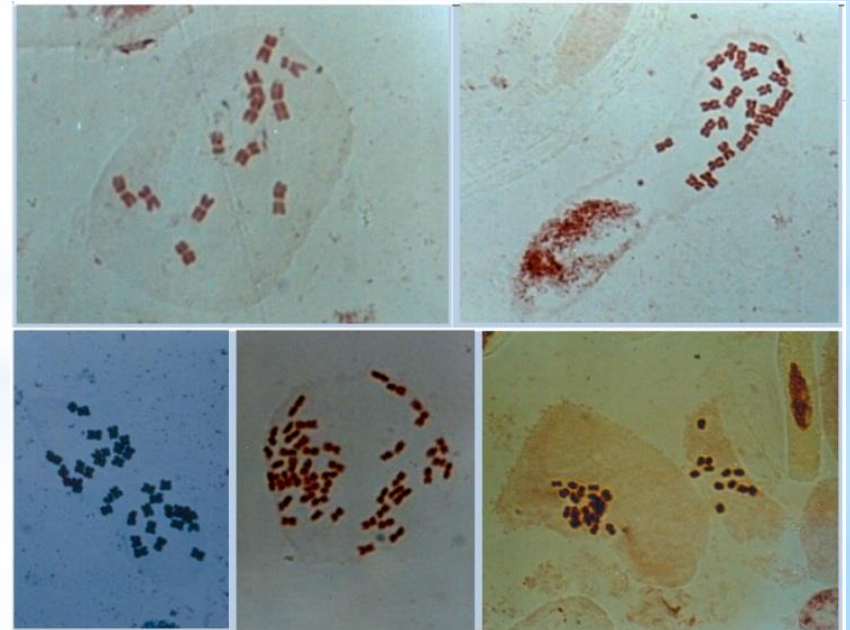




## \* 3. Genetic stability

### \* Patter of somatic mutation:

- \* Ploidy mutation : DNA endoreduplication),
- \* Floral mutation
  - \* Peloric mutant,
  - \* Labellum-like lateral petals,
  - \* Anthe-like lateral petals,
  - \* Anthocyanin pattern mutation



**Table 2. Ploidy level in different organs of Dtps. Wedding promenade**

| Organ          | Nuclear DNA content(channel number) and<br>Number of Nuclei(%) |       |       |       |       |
|----------------|----------------------------------------------------------------|-------|-------|-------|-------|
|                | ~20                                                            | ~40   | ~80   | ~160  | ~320  |
|                | 1.5C                                                           | 3C    | 6C    | 12C   | 24C   |
| Pollen         | 76.82                                                          | —     | —     | —     | —     |
| Axillaries bud | —                                                              | 55.51 | 23.11 | 4.79  | —     |
| Leaves         | —                                                              | —     | 11.7  | 28.64 | 27.12 |

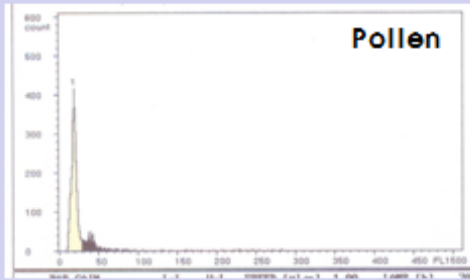
**Table 3. Ploidy level in different organs of P. Yellow Star**

| Organ          | Nuclear DNA content(channel number) and<br>Number of Nuclei(%) |       |       |       |      |
|----------------|----------------------------------------------------------------|-------|-------|-------|------|
|                | ~30                                                            | ~60   | ~120  | ~240  | ~360 |
|                | 2C                                                             | 4C    | 8C    | 16C   | 32C  |
| Pollen         | —                                                              | 52.13 | —     | —     | —    |
| Axillaries bud | —                                                              | —     | 39.71 | 20.37 | —    |
| Leaves         | —                                                              | —     | 6.92  | 24.69 | —    |

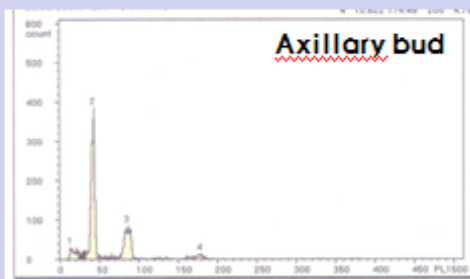


## Wedding <sup>N</sup>promenade

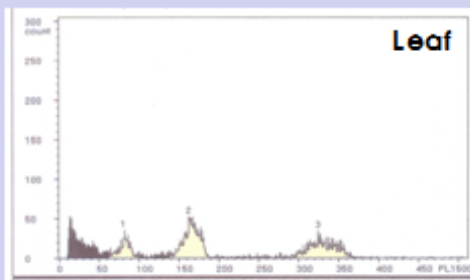
Pollen



Axillary bud



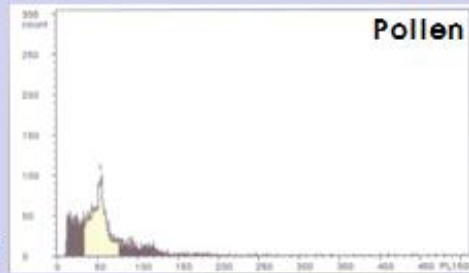
Leaf



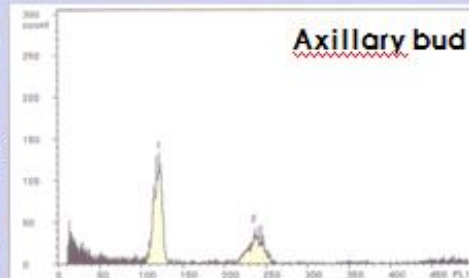
Nuclear DNA Content

## Yellow Star

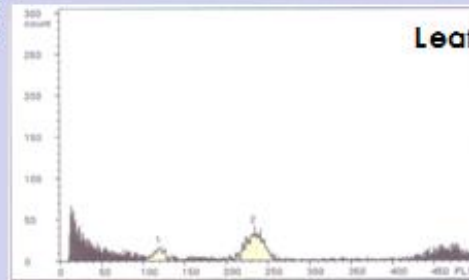
Pollen



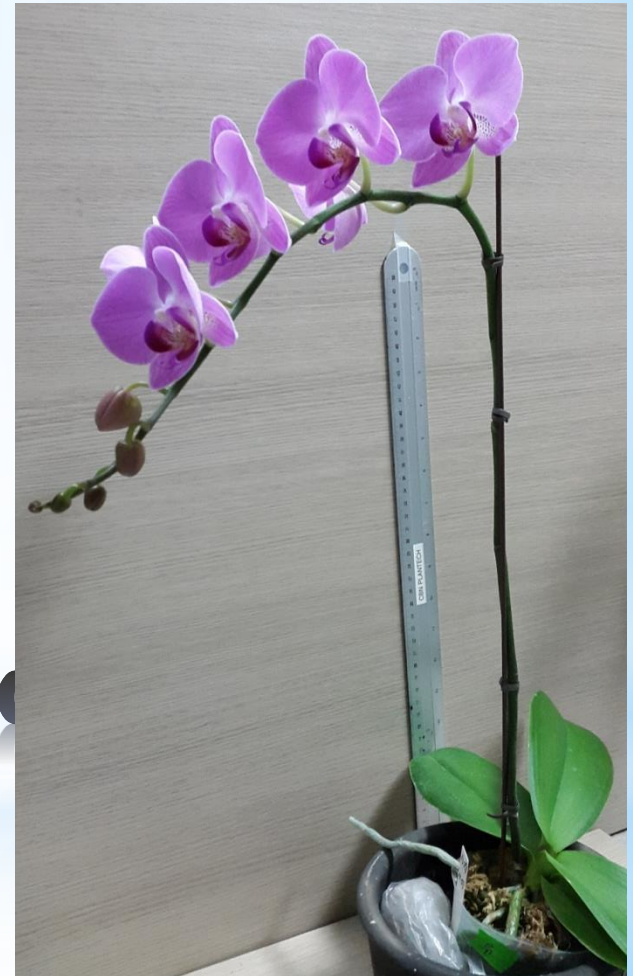
Axillary bud



Leaf



Nuclear DNA Content



# \* Peloric mutant Labellum-like lateral petals



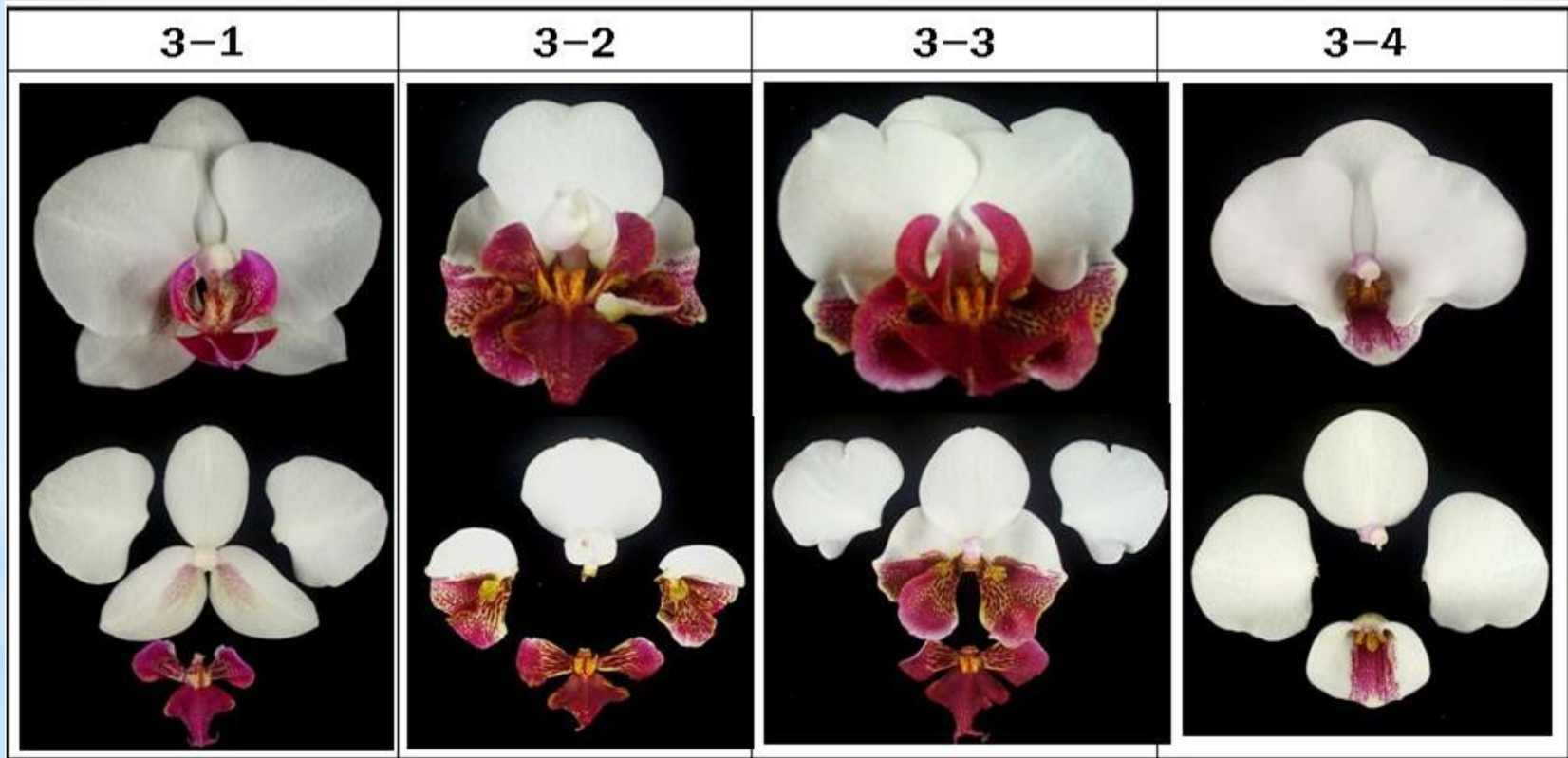
Normal



AbNormal



# \* Peloric mutant Labellum-like lateral petals



\*\*Orchid flowers with abnormal numbers of petals or lips are called peloric

# \* Peloric mutant Labellum-like lateral petals

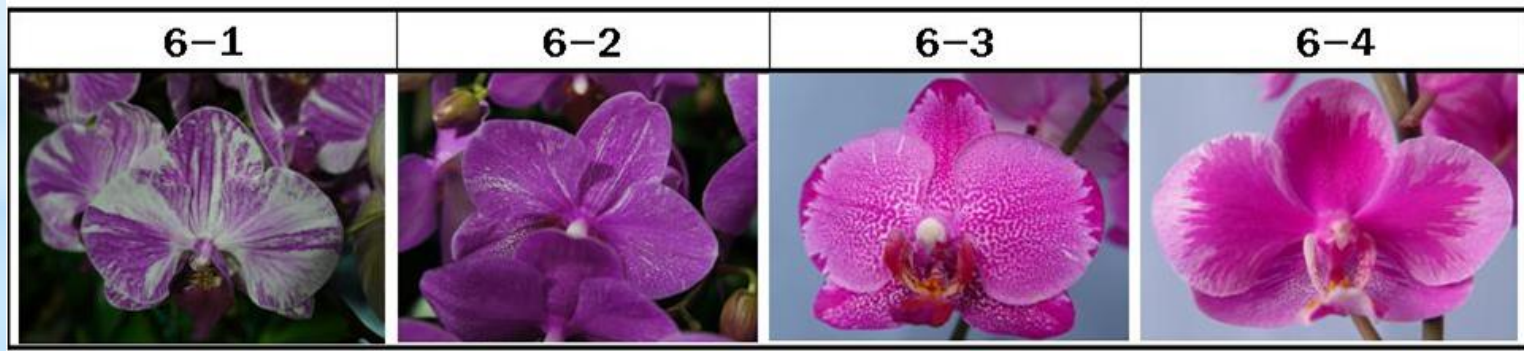




# \* Anther-like lateral petals



## Anthocyanin pattern mutation



# \* Petal color variation







## \* B. Breeding technique “Cross breeding”

- \* Self and cross pollination
  - \* Put the pollinia into the stigma
  - \* The time - Just blooming
  - \* Pollinating 2~3 flowers and remaining 1 healthy pod
- \* Harvest : 4~5 months after pollination
- \* Sowing : Sown the seeds on the flask

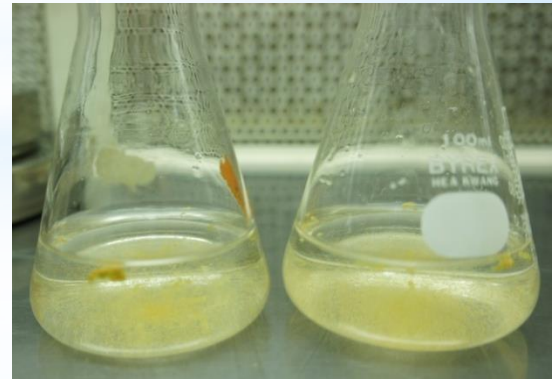
# \* Sterilization and seedling

## Green pod

1. Surface cleaning with 70% ethanol→
2. soak and shake in 4% sodium hypochloride during 15min→
3. 4 times rinse with Distilled water

## Open pod

1. Put in 70% ethanol during 20sec→ 5min Centrifuge at 1000rpm
2. Put in 2% sodium hypochloride during 5min→ 5min Centrifuge at 1000rpm
3. 3times rinse with distilled water(every time doing centrifuge)



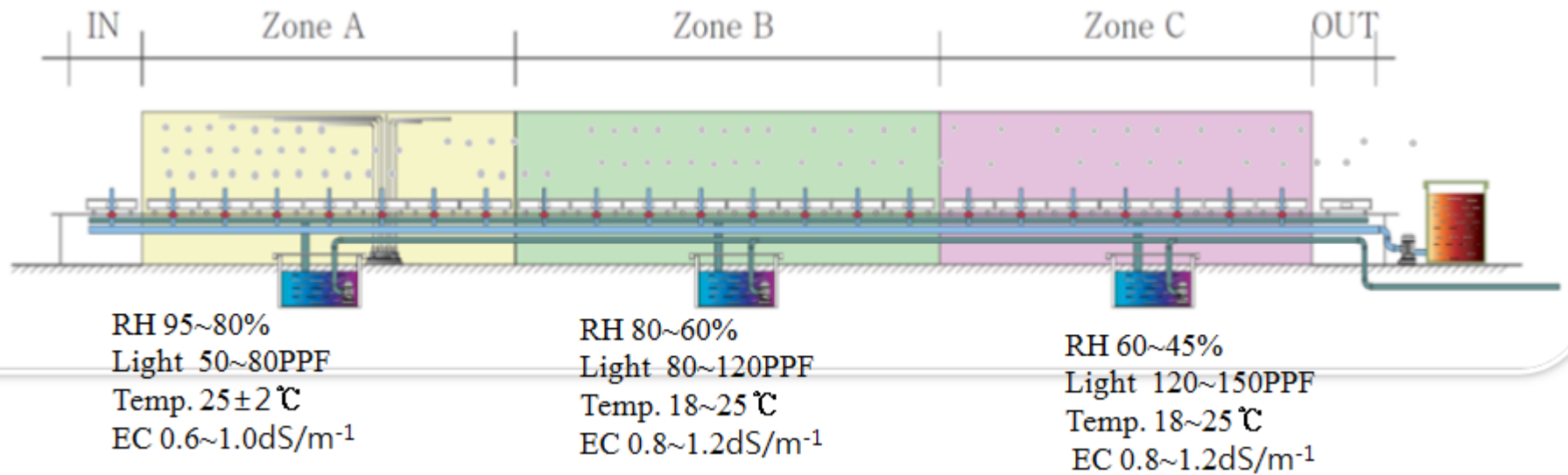


# \* Sterilization and seedling

- \* The PLB cluster germinated 2-3 months after sowing
- \* It is sub-cultured in a new medium
- \* At first transplanting, it is transplanted 2 or 3 times
- \* About 10 months after sowing can obtain health young in vitro plants



# \* Acclimatization (Microponic system)

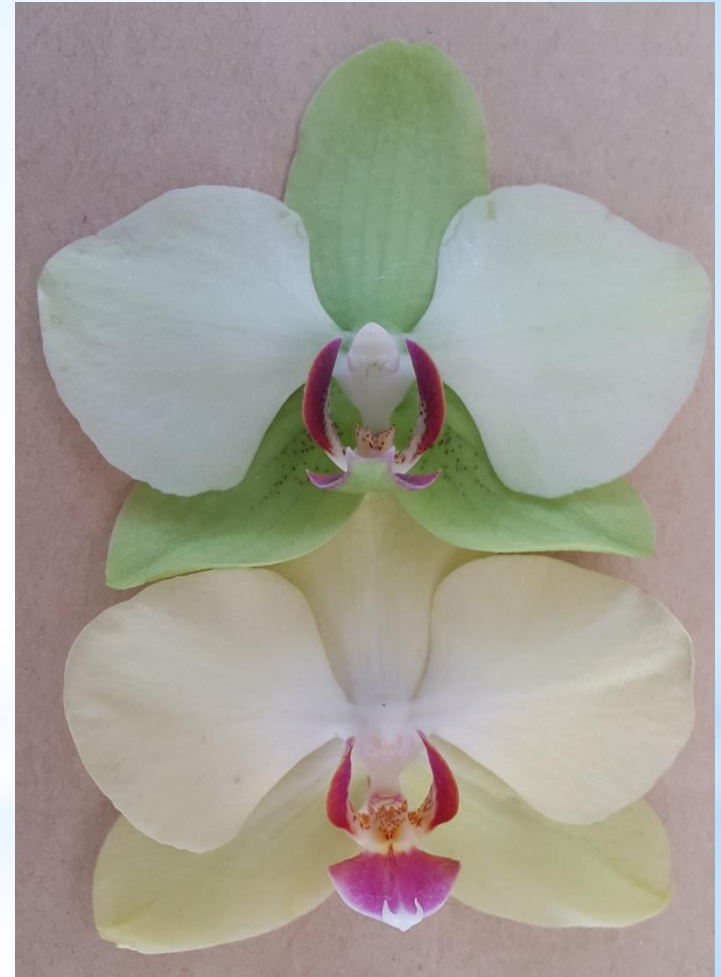


# \* Progressing of the breeding (Cross breeding)

|                         |                                                                                     |                                           |
|-------------------------|-------------------------------------------------------------------------------------|-------------------------------------------|
| Process of breeding     |    | Crossing                                  |
| 4 months                |    | sowing/IN VITRO                           |
| 10 months               |    | Culture IN VITRO                          |
| 24 months               |    | Cultivating and 1 <sup>st</sup> selection |
| 24months                |    | Clone propagation from seed               |
| 24months                |    | Selection & test                          |
| 12months                |  | Propagation from clone                    |
| 24months                |  | Mass propagation                          |
| Total 122moths,10years) |  | New varieties                             |



# \*Selection & test





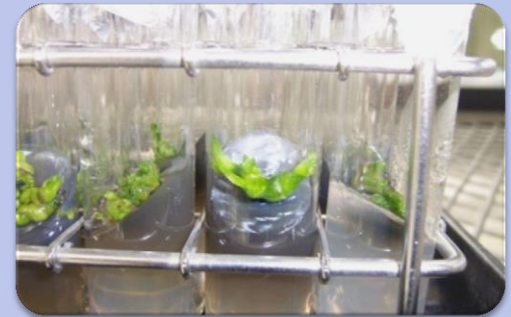








# \* Selection & in vitro propagation





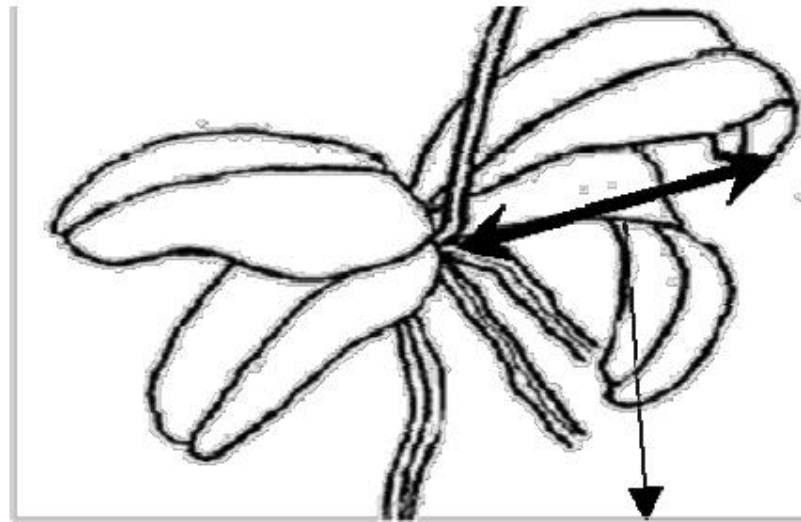


# \* UPOV Test Guideline for Phalaenopsis

## 1. Plant size

〈부표〉 특성표 설명

특성 1. 식물체 : 크기



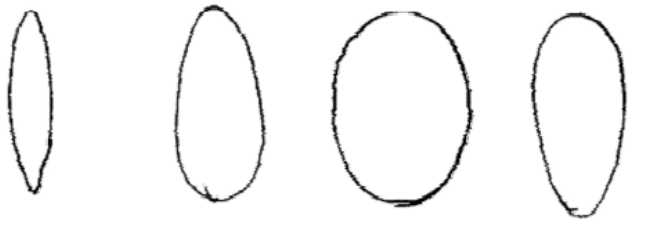
축의 하부에서 최장엽의 끝까지



# \* UPOV Test Guideline for Phalaenopsis

- \* 2~9. Leaf : length, with, shape, shape of apex, symmetry of apex, attitude, color of upper side, anthocyanin coloration,

특성 4. 잎 : 모양



1. 선형      2. 좁은난형      3. 타원형      4. 좁은 도란형

특성 5. 잎 : 선단의 모양



1. 예형      2. 둔형      3. 요형

특성 6. 잎 : 선단의 대칭



1. 비대칭      2. 대칭

특성 7. 잎 : 자세



3. 반직립      5. 수평      7. 반하수

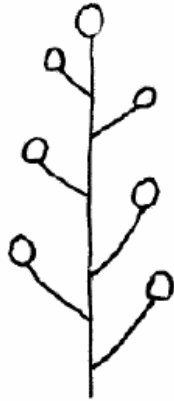
# \* UPOV Test Guideline for Phalaenopsis

\* 10~12. Inflorescence : type, length, number of flowers,

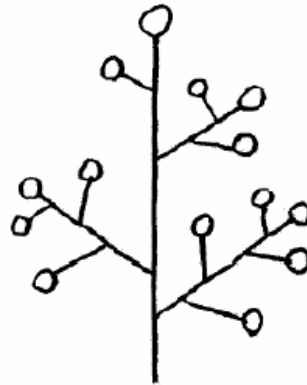
특성 10. 화서 : 형태



1. 단정화서



2. 총상화서



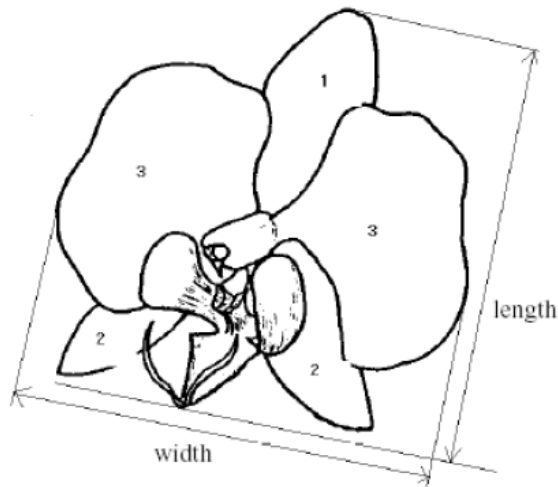
3. 복총상화서

특성 11. 화서 : 길이

# \* UPOV Test Guideline for Phalaenopsis

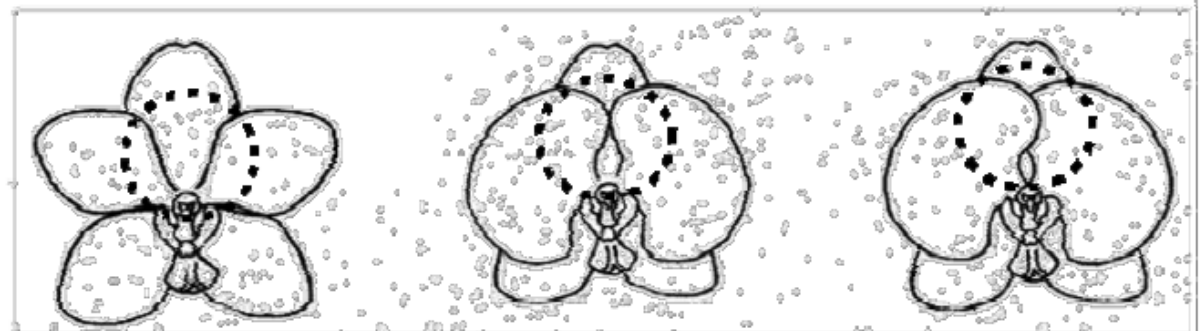
특성 18, 19. 꽃 : 정면에서 측정한 길이, 너비

1. 윗 꽃받침
2. 측면꽃받침
3. 꽃잎



- \* 13~15. Peduncle : length, thickness, anthocyanin coloration
- \* 16~21. Flower : general appearance of petals and sepals, texture of the surface of sepals and petals, length in front view, width in front view, arrangement of petals, fragrance

특성 20. 꽃 : 꽃잎의 배열



1. 떨어짐

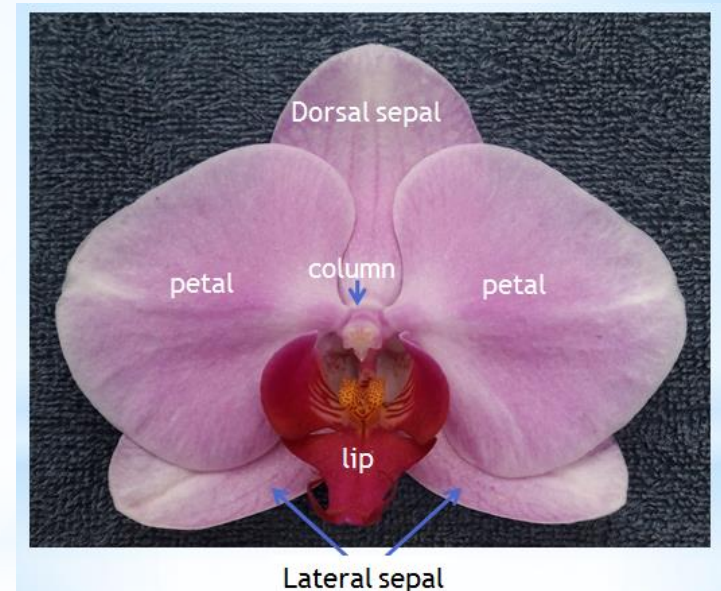
2. 접촉

3. 겹침



# \*UPOV Test Guideline for Phalaenopsis

- \* 22~28. Sepal : shape, length, width, curvature of longitudinal axis, cross section, twisting, undulation of margin
- \* 29~32. Dorsal sepal : colors pattern, main colors, secondary colors
- \* 33~36. Lateral sepal : number of colors, colors pattern, main colors, secondary colors
- \* 37~46. Petal : shape, length, with, curvature of longitudinal axis, shape in cross section, twisting, undulation of margin, number of colours, colour pattern, main colour

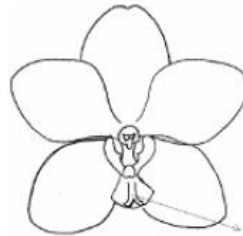
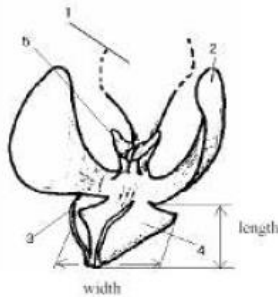


# \* UPOV Test Guideline for Phalaenopsis

- \* 47~66. Lip : length of apical lobe, with of apical lobe, presence of whiskers, length of whiskers relative to length of apical lobe, shape of apical lobe, bump and ridge on apical lobe, type of curvature of lateral lobe, size of lateral lobe relative to apical lobe, number of colours, colour pattern of apical lobe, main colour of apical lobe, colour of pattern of apical lobe, colour pattern of lateral lobe, main colour of lateral lobe, colour of pattern of lateral lobe, callus , pubescence

특성 49, 50, 52, 54. 순판 : 끝 열편의 길이, 너비, 수염의 길이, 끝열편 돌기 및 융기

1. 꽃술대
2. 순판 : 측열편
3. 순판 : 수염
4. 순판 : 끝열편
5. 순판 : 캘투스



돌기 및 융기

특성 53. 순판 : 끝 열편의 모양



1. 난형



2. 타원형



3. 도란형



4. 원형



5. 반원형



6. 삼각형



7. 역삼각형

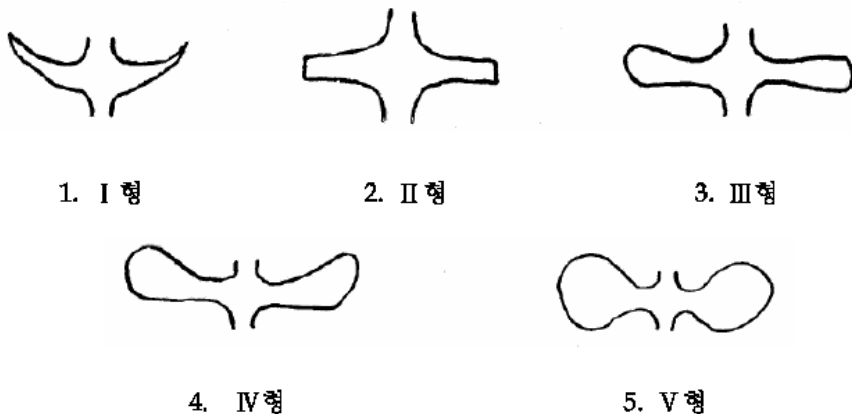


8. 마름모형

# \* UPOV Test Guideline for Phalaenopsis

- \* 55~66. Lip : type of shape of lateral lobe, type of curvature of lateral lobe, size of lateral lobe relative to apical lobe, number of colours, colour pattern of apical lobe, main colour of apical lobe, colour of pattern of apical lobe, colour pattern of lateral lobe, main colour of lateral lobe, colour of pattern of lateral lobe, callus , pubescence
- \* 67. Column : colour of apex

특성 55. 순판 : 측열편의 유형



특성 56. 순판 : 측열편의 만곡 형태





### 3. New varieties breeding by UNiPlantech

